

Data Center Impact Explorer

G1 - JAMA Women

**Data Science Capstone Project
Launch Report**

Date:

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Team Members:

Name: Andrea MacGregor

Name: Anna Grace

Name: Jillian Kunze

Name: Morgan Murphy

The System/Product

System/Product Name: Data Center Impact Explorer

Introduction:

Data centers are the backbone of the modern Internet and provide the necessary physical storage space to enable cloud storage, digital streaming, and artificial intelligence applications (Malone, 2026). It is the latter that has been causing a particularly explosive growth in the construction of data centers across the U.S. in recent years (Stansbury et al., 2025). Yet that growth comes at a cost, and a groundswell of community opposition has raised concerns about the impact of data center construction and operation at the local level (Jones, 2026). Data centers require significant amounts of water and energy, often from natural gas, and can cause noticeable noise pollution. The centers can have a somewhat positive impact on the local economy, but the creation of stable, long-term jobs at these sites is often low (Malone, 2026).

Despite the growing public concern, concrete information about the impacts of data centers can be difficult to find and is spread across many different sites on the web. Our goal is to bring together some of these disparate sources of information to gain a better understanding of the relationship between data center siting, local and regional politics, and health and environmental effects. We will perform exploratory data analysis and use statistical techniques to learn more about these connections, and then present them via data visualization.

Our final deliverable will be an interactive webpage with visualizations pertaining to data centers and their impacts. We plan to show a map of data center locations, and then a breakdown of the relationships with other attributes of those locations - political leanings, local power supplies, water and air quality, and mental and physical health. These interactive visualizations will show the results of our data aggregation and analysis in an easy-to-interpret manner. By exploring multiple aspects of the interplay between data centers and local communities, we hope to gain a more holistic understanding of the complex relationships between this rapidly expanding industry and the people who are most impacted.

Highlighted Features:

- Data joining and aggregation of multiple disparate datasets, bringing together information that is currently spread across the Internet in various forms into one easy-to-access location
- Investigation of questions on the incidence and impact of data centers:
 - Where are data centers predominantly located across the U.S.?
 - Are data centers built more often in areas that are lean more Democratic or Republican? How might that affect public sentiment or community opposition?
 - In what types of environments are data centers often situated? How are data centers impacting air quality, water resources, and power usage?
 - How do data centers impact the physical and mental health of people in the surrounding community?
- Accessible, interactive visualization and explanation of key findings and insights on a web platform

Sponsor or Proxy User:

Our team identified environmental advocate Erin Brockovich as a potential sponsor or user because of her longstanding work investigating the environmental and public health impacts of industrial development on local communities, and her current efforts to track data center development via community reporting (Brockovich AI Data Center Reporting, 2026). While we have reached out, we have not yet received a response.

Issues:

The primary challenge for this project is data acquisition and integration. Information related to data centers is distributed across numerous federal, state, and private sources. These datasets vary in quality, making them difficult to parse and combine into a single, consistent dataset. Additionally, some information like facility-level resource consumption is incomplete or unavailable due to proprietary restrictions.

Another challenge will be interpreting relationships between data center development and environmental or health outcomes. While our analysis may identify statistical associations, establishing direct causation is outside of the scope of this project. By clearly documenting data limitations and analytical assumptions, we aim to present results that are transparent and informative.

The Team

Team Name: G1 - JAMA Women

Team Members and their Specialties:

Since our individual team backgrounds vary greatly, each of our members has unique and diverse specialties. Going in the order of our team name acronym, our backgrounds are as follows:

- **Jillian:** Jillian has an academic background in physics and astronomy, as well as professional experience in science communication. Recently, she has completed course-related projects involving the use of social network analytics, time-series analysis methods, and recommender systems. Her largest strengths in a team setting are her exceptional literary craftsmanship and comprehensive approach to exploratory data analysis. Jillian previously completed a project related to frontier AI data centers in an Information Visualization class; so, she already has some knowledge of relevant issues, but will use different data for different purposes in this project. She is interested in data related to government happenings (including local, federal, and global institutions), how they may impact the involved communities, and how different analysis methods of such data can contribute to a more well-rounded and well-informed general public. For this project, she will be examining the political structures that underlie the siting of data centers (for example, whether they are more likely to be built in red or blue districts in the U.S.).
- **Anna:** Anna has an academic background in computer science and mathematics, with professional experience in financial analysis. She has recently completed course-related projects involving clustering methods, recommender system processes, and database management

procedures. Her greatest strengths when in a team are her extensive Python knowledge (further supported by her comprehensive understanding of the complex math working behind the scenes) and ability to use her knowledge of the data science life cycle to inform project steps upfront. She is particularly interested in using data to investigate complex real-world problems with environmental and societal impacts. For this project, she will focus on exploring the impacts of data centers on local power supplies.

- **Morgan:** Morgan has an academic background in psychology and human factors engineering, with professional experience in medical technology usability research. She has recently completed course-related projects involving A/B testing, clustering techniques, and social network analytics. Her largest strengths in a team are her ability to identify and analyze literature that both supports and justifies the proposed project and her knowledge of effective data visualization techniques. She is interested in data broadly related to the health sciences sector—particularly involving burgeoning medical technology and the access various populations have to healthcare—as well as the publishing industry. For this project, she will be attempting to determine if there is a connection between data centers and various health issues/diagnoses.
- **Andrea:** Andrea has an academic background in engineering and professional experience with business intelligence analysis. She has recently completed course-related projects involving time-series analysis methods, A/B testing, and usability best practices. In a team, her greatest strengths are her ability to clearly and concisely explain her process(es) as well as her outstanding ability to create interactive dashboards with tools such as Power BI and Tableau. She is interested in data involving the environment, including national parks and various metrics related to environmental health. For this project, she will be investigating the environmental impact of data centers, including aspects such as air, soil, and/or water quality.

Additionally, all team members have recent academic experience with tools and processes relevant to the data science life cycle, including (but not limited to) preprocessing and data cleaning techniques, AI/ML models, and data/statistical analysis methods. While some individuals are more experienced with Python, all team members have successfully and repeatedly utilized the language to complete data science projects and coursework. As such, all will contribute to data acquisition, merging, and pre-processing, since that will be such a major component of this project.

Team Communication:

We will use Microsoft Teams and text for day-to-day communication and file sharing, while maintaining a shared Google Drive folder for datasets, documentation, and collaborative work. Google Colab can be used for data cleaning and exploratory data analysis. While each team member's work should be separate, if we find that we need version management or more control over overlapping code, we will publish our work on GitHub. We plan for our meetings to be during the off weeks of our DSCI 591 class, when we do not meet in person on Tuesdays from 6 - 9 pm. We also have time for impromptu meetings on Thursdays from 5:30 - 6:30 pm. In between our meeting times, we will communicate via messaging.

Team Issues:

An issue that we face as Master's students is that we all have various life commitments outside of school that make it hard for our team to meet and work together throughout the week. To combat this, we will use the time that we have blocked out for this class in our schedule to meet when we do not have class in

person. This will allow us to meet on a regular basis at a time that we know we have available in our schedules. This will also allow us to keep the project moving forward, even when we don't have in-class deliverables, and help us stay in constant communication.

Another potential challenge is that each team member comes from different backgrounds and has different strengths and weaknesses. Our team can use this to our advantage by having our members work on parts of the project that they enjoy or are strong in. We can also expand our individual skillsets by working together and learning from one another. Our different skill sets can be an asset if used correctly.

Project Plan

Project Timeline:

Phase	Activities	Deliverables
1. Team formation and Project Definition (Week 1)	Finalize project topic, identify research questions, assign team roles, and determine potential data sources related to data centers, environmental quality, public health, and demographics. Contact potential sponsor.	Team formation
2. Data Source Identification (Week 2)	Locate and evaluate datasets (e.g. data center locations, EPA environmental data, Census demographics, health indicators). Document data availability, geographic resolution, and licensing.	Launch report; dataset inventory and data source documentation
3. Data Acquisition & Understanding (Week 3)	Download datasets, examine variables, and identify common geographic identifiers. Create a data dictionary.	Launch presentation; dataset description and data dictionary
4. Data Combination, Cleaning, and Preprocessing (Week 4)	Clean individual datasets, standardize geographic identifiers, handle missing values, and merge datasets into a unified analysis dataset.	Cleaned, integrated dataset and preprocessing code
5. Project Pitch Preparation (Week 5)	Prepare a pitch presentation for class identifying project motivations, available datasets, preliminary findings, proposed analyses, and interactive dashboard design.	Pitch presentation
6. Exploratory and Statistical Analysis (Week 6 - 7)	Explore geographic patterns, analyze relationships with environmental quality, political affiliation, water usage, air quality,	Data acquisition and pre-processing report;

	and health metrics using descriptive statistics and hypothesis testing where appropriate.	data analysis and statistical analysis code
7. Interactive Visualization Development (Week 8 - 9)	Build interactive maps and dashboards showing data center locations alongside environmental, demographic, political, and health indicators. Summarize key insights and trends for users.	Interactive web application, visualizations, and analysis report
8. Final Documentation & Presentation (Week 10)	Refine analyses, finalize the interactive website, complete written documentation, and prepare the final presentation and demonstration.	Final report, interactive website, and final presentation

Relevant Questions:

1. What problem are we solving?

While various social justice groups are working to increase transparency surrounding data centers and the implications they have on the communities they were built in, they usually focus on one area of impact, leading to the presence of numerous siloed resources. With this project (supported by our interactive site), we are aiming to create a more comprehensive resource that displays the wide variety of repercussions a data center can have on the local population. It is our hope that this can, in turn, assist the general population and policymakers alike in developing a thorough understanding of the weight this issue holds in regard to public wellbeing.

2. What data will we use?

We will be using publicly available, real-world data from various repositories. Major, overarching sources include Kaggle, the U.S. Data Centers Tracker from FracTracker Alliance (FracTracker, 2026), Brockovich AI Data Center Reporting (Brockovich, 2026), and the U.S. Drought Monitor (U.S. Drought Monitor, n.d.). Specific datasets will be reported in greater detail via later reports once they have been successfully identified and compiled.

3. What methods will we apply?

We will apply data cleaning, preprocessing, and integration techniques to combine multiple public datasets into a unified analytical dataset. We will then use exploratory data analysis, descriptive statistics, correlation analysis, and hypothesis testing to investigate relationships between data center locations and environmental, political, and health factors. Finally, we will present our findings through interactive maps and visualizations that allow users to explore these relationships.

4. How will we measure success?

Success will be determined through three key values, which also correlate to the life cycle of this project:

- **Variety:** Do we feel that we have sufficiently demonstrated the wide range of effects a data center can have on the community and environment it was built in? Would the presentation of this information make an individual more informed than they were before on the topic?
- **Integrity:** Have we adequately preprocessed the data, including factors such as compiling/combining different datasets correctly and conducting the appropriate cleaning steps? Is our data reputable, reliable, and aligned with our purpose?
- **Communicability:** Have we presented our information in an approachable, interpretable way? Can the intended audience implicitly understand the point we are making without too much difficulty?

We will intermittently return to these values throughout the development of our project to ensure that we are on track to successfully achieve our goals.

5. *Who does what and when?*

All team members will abide by the values listed above for each of their respective areas of interest. Furthermore, all team members will contribute to each phase as outlined in our Project Timeline within these areas; contributions will be divided up in this manner as opposed to each member being assigned a specific step (such as data acquisition or cleaning) within the broader timeline.

Table of Contributions

The table below identifies contributors to various sections of this document.

	Section	Writing	Editing
1	Project	Jillian (Wrote Introduction and Highlighted Features sections) Anna (Wrote Sponsor or Proxy User and Issues sections)	Andrea (Reviewed and formatted) Morgan (Reviewed)
2	Team	Morgan (Wrote Team Name and Team Members and their Specialties sections) Andrea (Wrote Team Communication and Team Issues sections)	Anna (Edited section) Jillian (Review and formatting)
3	Plan	Andrea (Wrote Project Timeline) Morgan (Answered Relevant Questions)	Anna (Edited timeline and relevant questions) Jillian (Review and formatting) Morgan (Timeline Review)

References

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